

LOAD TABLE **STEEL GRATING**

LOAD TABLE FOR STEEL GRATING - TYPE W-19 OR P-19

ASTM A 1011 CS TYPE B
F=18,000psi, E=29,000,000psi

See Appendix A for a graphic depicting table loading

Recommended max. span (in.) for 1/4 in. deflection under uniform load of 100psf

U=uniform load, psf
D=deflection, in.
C=concentrated load at mid-span, lb per foot of grating width

Bearing Bar Size (in)	Nominal Weight (psf)**			Span in Inches									
				24	30	36	42	48	54	60	66	72	
3/4x1/8 [4]	42	U		355	227	158	116	89	70				
		Du		0.099	0.155	0.223	0.304	0.397	0.503				
		C		355	284	237	203	178	158				
		Dc		0.079	0.124	0.179	0.243	0.318	0.402				
3/4x3/16 [6]	46	U		533	341	237	174	133	105				
		Du		0.099	0.155	0.223	0.304	0.397	0.503				
		C		533	426	355	305	266	237				
		Dc		0.079	0.124	0.179	0.243	0.318	0.402				
1x1/8 [6]	51	U		632	404	281	206	158	125	101	84	70	
		Du		0.074	0.116	0.168	0.228	0.298	0.377	0.466	0.563	0.670	
		C		632	505	421	361	316	281	253	230	211	
		Dc		0.060	0.093	0.134	0.182	0.238	0.302	0.372	0.451	0.536	
1x3/16 [8]	57	U		947	606	421	309	237	187	152	125	105	
		Du		0.074	0.116	0.168	0.228	0.298	0.377	0.466	0.563	0.670	
		C		947	758	632	541	474	421	379	344	316	
		Dc		0.060	0.093	0.134	0.182	0.238	0.302	0.372	0.451	0.536	
1-1/4x1/8 [7]	61	U		987	632	439	322	247	195	158	130	110	93
		Du		0.060	0.093	0.134	0.182	0.238	0.302	0.372	0.451	0.536	0.629
		C		987	789	658	564	493	439	395	359	329	304
		Dc		0.048	0.074	0.107	0.146	0.191	0.241	0.298	0.360	0.429	0.504
1-1/4x3/16 [9]	67	U		1480	947	658	483	370	292	237	196	164	140
		Du		0.060	0.093	0.134	0.182	0.238	0.302	0.372	0.451	0.536	0.629
		C		1480	1184	987	846	740	658	592	538	493	455
		Dc		0.048	0.074	0.107	0.146	0.191	0.241	0.298	0.360	0.429	0.504
1-1/2x1/8 [8]	70	U		1421	909	632	464	355	281	227	188	158	135
		Du		0.050	0.078	0.112	0.152	0.199	0.251	0.310	0.376	0.447	0.524
		C		1421	1137	947	812	711	632	568	517	474	437
		Dc		0.040	0.062	0.089	0.122	0.159	0.201	0.248	0.300	0.358	0.420
1-1/2x3/16 [11]	77	U		2132	1364	947	696	533	421	341	282	237	202
		Du		0.050	0.078	0.112	0.152	0.199	0.251	0.310	0.376	0.447	0.524
		C		2132	1705	1421	1218	1066	947	853	775	711	656
		Dc		0.040	0.062	0.089	0.122	0.159	0.201	0.248	0.300	0.358	0.420
1-3/4x3/16 [13]	87	U		2901	1857	1289	947	725	573	464	384	322	275
		Du		0.043	0.067	0.096	0.130	0.170	0.215	0.266	0.322	0.383	0.450
		C		2901	2321	1934	1658	1451	1289	1161	1055	967	893
		Dc		0.034	0.053	0.077	0.104	0.136	0.172	0.213	0.257	0.306	0.360
2x3/16 [14]	96	U		3789	2425	1684	1237	947	749	606	501	421	359
		Du		0.037	0.058	0.084	0.114	0.149	0.189	0.233	0.282	0.335	0.393
		C		3789	3032	2526	2165	1895	1684	1516	1378	1263	1166
		Dc		0.030	0.047	0.067	0.091	0.119	0.151	0.186	0.225	0.268	0.315
2-1/4x3/16 [16]	105	U		4796	3069	2132	1566	1199	947	767	634	533	454
		Du		0.033	0.052	0.074	0.101	0.132	0.168	0.207	0.250	0.298	0.350
		C		4796	3837	3197	2741	2398	2132	1918	1744	1599	1476
		Dc		0.026	0.041	0.060	0.081	0.106	0.134	0.166	0.200	0.238	0.280
2-1/2x3/16 [18]	113	U		5921	3789	2632	1933	1480	1170	947	783	658	561
		Du		0.030	0.047	0.067	0.091	0.119	0.151	0.186	0.225	0.268	0.315
		C		5921	4737	3947	3383	2961	2632	2368	2153	1974	1822
		Dc		0.024	0.037	0.054	0.073	0.095	0.121	0.149	0.180	0.215	0.252

Conversion Factors:
For gratings with other than 1-3/16" bearing bar spacing, or for different design stresses, proportionate conversion factors apply. Refer to the Metal Bar Grating Engineering Design Manual for the development of such factors.

Note: 1/4" is considered the maximum deflection consistent with pedestrian comfort, but can be exceeded for other loading conditions at the discretion of the engineer.

NOTE: For serrated grating, the depth of grating required for a specified load is 1/4" greater than in the table.

**Weights (mass/area) shown are approximate and vary with manufacturers. They are provided for preliminary design computations only and are not intended for any other purpose.

Loads above DO NOT include the dead load of the grating.